File

Name: R5-fit-A4-T500
Description: KISSsoft example
Changed by: AAA on: 31.10.2026 at: 04:08:34

1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00 /10.00			
Diameter hub (mm)	[DaA]/[DaI]	125.00 /50.00			
Length of Interference fit (mm)	[l]	40.00			
Diameter of joint (mm)	[DF]	50.00			
with special allowances:					
Upper allowance Shaft (µm)	[AoI]	65.0			
Lower allowance Shaft (µm)	[AuI]	65.0			
Tolerance measure Shaft (µm)	[TI]	0.0			
Upper allowance Hub (µm)	[AoA]	0.0			
Lower allowance Hub (µm)	[AuA]	0.0			
Tolerance measure Hub (µm)	[TA]	0.0			
Largest interference (µm)	[Po]	65.0			
Smallest interference (µm)	[Pu]	65.0			
Nominal torque (Nm)	[T]	500.00			
Application factor	[KA]	1.00			
Service torque (Nm)	[Tb]	500.00			
Axial force (N)	[FA]	0.00			
Bending moment (Nm)	[Mb]	0.00			
Radial force (N)	[Fr]	0.00			
Circumferential force (N)	[Fu]	20000.00			
Speed (1/min)	[n]	0.00			
Interference (µm)	[P]	65.0	(65.0	..	65.0)
Embedding (µm)	[s]	4.80			
Effective interference at 20°C(68°F) (µm)	[Pw]	60.20	(60.2	..	60.2)
Effective interference at working temp.. (µm)	[PwTh]	60.20	(60.2	..	60.2)
Service temperature shaft (°C)	[ThB]	20.0			
Service temperature hub (°C)	[ThB]	20.0			
Pressure stress by					
- Interference (after mounting) (N/mm²)	[pM]	43.53	(43.53	..	43.53)
- Interference (working) (N/mm²)	[p]	43.53	(43.53	..	43.53)
- Bending moment (N/mm²)	[pb]	0.00			
- Radial force (N/mm²)	[pr]	0.00			
Coefficient of friction, longitudinal	[µa]	0.120			
Coefficient of friction, circumferential	[µu]	0.120			
Safety against sliding	[Sr]	1.64	(1.64	..	1.64)
Required safety against sliding	[SSr]	1.50			

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	650.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The shaft's material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	21			
Surface class of roughness	Own Input				
Surface roughness (µm)	[Rz]	6.00			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-7.93	(-7.93	..	-7.93)
Diameter increase during operation (o/oo)	[ΔD]	-0.16	(-0.16	..	-0.16)
Equivalent stress outer diameter (N/mm²)	[σVa]	45.46	(45.46	..	45.46)
- Radial stress (N/mm²)	[σRa]	-43.53	(-43.53	..	-43.53)
- Tangential stress (N/mm²)	[σTa]	-47.16	(-47.16	..	-47.16)
- With outside load (N/mm²)	[σVaMF]	45.46	(45.46	..	45.46)
Inner diameter (mm)	[Dil]	10.00			
Diameter increase during operation (µm)	[ΔD]	-4.22	(-4.22	..	-4.22)
Diameter increase during operation (o/oo)	[ΔD]	-0.42	(-0.42	..	-0.42)
Equivalent stress inner diameter (N/mm²)	[σVi]	90.69	(90.69	..	90.69)
- Radial stress (N/mm²)	[σRi]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTi]	-90.69	(-90.69	..	-90.69)
Safety against fracture	[SiRm]	7.17	(7.17	..	7.17)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	3.31	(3.31	..	3.31)
Required safety against yield point	[SSi.Rp]	1.20			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	70000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	150.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	40			
Surface class of roughness	Own Input				
Surface roughness (µm)	[Rz]	6.00			
External diameter (mm)	[DaA]	125.00			
Diameter increase during operation (µm)	[ΔD]	29.61	(29.61	..	29.61)
Diameter increase during operation (o/oo)	[ΔD]	0.24	(0.24	..	0.24)
Equivalent stress outer diameter (N/mm²)	[σVa]	16.58	(16.58	..	16.58)
- Radial stress (N/mm²)	[σRa]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTa]	16.58	(16.58	..	16.58)
Inner diameter (mm)	[Dal]	50.00			
Diameter increase during operation (µm)	[ΔD]	52.27	(52.27	..	52.27)
Diameter increase during operation (o/oo)	[ΔD]	1.05	(1.05	..	1.05)
Equivalent stress inner diameter (N/mm²)	[σVi]	90.15	(90.15	..	90.15)
- Radial stress (N/mm²)	[σRi]	-43.53	(-43.53	..	-43.53)
- Tangential stress (N/mm²)	[σTi]	60.12	(60.12	..	60.12)

- Radial stress with outside load (N/mm²)	[σRai]	-43.53	(-43.53	..	-43.53	)
- Tangential stress with outside load (N/mm²)	[σTai]	60.12	(60.12	..	60.12	)
- With outside load (N/mm²)	[σViMF]	90.15	(90.15	..	90.15	)
Safety against fracture	[Si.Rm]	7.77	(7.77	..	7.77	)
Required safety against fracture	[SSi.Rm]	1.50				
Safety against yield point	[Si.Re]	1.66	(1.66	..	1.66	)
Required safety against yield point	[SSi.Re]	1.20				

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:						
Mounting clearance (mm)	[PsTh]	0.050				
Temperature difference for mounting:						
Shaft temperature: (°C) Hub temperature: [ThA] (°C)						
Mount with shaft temperature 20 °C:						
Assembly temperature shaft (°C)	[θI]	20				
Assembly temperature hub (°C)	[θA]	220.000				
Mount with shaft temperature -150 °C:						
Assembly temperature shaft (°C)	[θI]	-150				
Assembly temperature hub (°C)	[θA]	94.348				
calculated using the coefficient of thermal expansion						
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50				
Longitudinal pressure fit:						
Assembly temperature shaft (°C)	[θM]	20.00				
Assembly temperature hub (°C)	[θM]	20.00				
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16			
Press on (force) (kN)	[Fpress]	42.67	(42.67	..	42.67	)
Coefficient of friction, longitudinal	[μll=μa*1.600]	0.19				
Press out (force) (kN)	[Fpress]	52.52	(52.52	..	52.52	)
Notice:						
Micro sliding can occur in Interference fit!						
Risk of friction corrosion, fretting.						
Coefficient of friction	[μ]	0.19				
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	265.87	(265.87	..	265.87	)

Notice concerning the display: Number-1 (Number-2.. Number-3):

Number-1: Value calculated with the mean allowance

Number-2: Value of the smallest possible allowance

Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 160)
